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SECTOR AT A GLANCE

Digital Health

Health Sector in India

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Digital Health

Health Sector in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

In recent years, India has witnessed a remarkable evolution and progress in the field of digital health. This document explores the transformative impact of digital health technologies on healthcare delivery, patient outcomes, and the healthcare ecosystem for India and also delves into the key highlights of various initiatives by the Government of India to boost and encourage the growth of digital health in the country.

According to WHO, Digital Health is defined as the “*field of knowledge and practice associated with the development and use of digital technologies to improve health. Digital health expands the concept of eHealth to include digital consumers, with a wider range of smart devices and connected equipment. The following areas are commonly understood as being part of, or related to, digital health: artificial intelligence, big data, blockchain, health data, health information systems, the infodemic, the Internet of Things(IoT), interoperability and telemedicine*”¹

Evolution of Digital Health in India



The journey of digital health started with the **National Health Policy (NHP) 2017** which advocates extensive deployment of digital tools for improving the efficiency and outcome of the healthcare system. The policy aims at an integrated health information system which serves the needs of all stakeholders and improves efficiency, transparency, and citizen experience². Delivery of better health outcomes in terms of access, quality, affordability, lowering of disease burden and efficient monitoring of health entitlements to citizens, is the goal.³



Policy supported to establish **federated national health information architecture**, to roll-out and link systems across public and private health providers at state and national levels consistent with Metadata and Data Standards (MDDS) and Electronic Health Record (EHR).²



The policy suggests exploring the use of “Aadhaar” (Unique ID) for identification. Creation of registries (i.e., patients, provider, service, diseases, document and event) for **enhanced public health/big data analytics, creation of health information exchange platform and national health information network, use of National Optical Fibre Network**, use of smartphones/tablets for capturing real time data, as the key strategies of the National Health Information Architecture.²



The policy advocates scaling of various initiatives in the area of **teleconsultation** which will entail linking tertiary care institutions (medical colleges) to District and Subdistrict hospitals which provide secondary care facilities, for the purpose of specialist consultations, promote utilization of National Knowledge Network for Tele-education, Tele-CME, Teleconsultations and access to digital library.²



Implementation

In a follow-up of the NHP's specific goals for **adopting digital technologies**, the Ministry of Health and Family Welfare constituted a committee headed by **Shri J. Satyanarayana** to develop an implementation framework for the **National Health Stack**. This committee produced the National Digital Health **Blueprint (NDHB)**², laying out the building blocks and an action plan to comprehensively and holistically implement digital health.

Subsequently, the National Digital Mission was revamped under the umbrella of Ayushman Bharat Digital Mission (ABDM)

1. World Health Organization.(n.d.). *Digital Health*. Retrieved 1 September 2023.

2. Ministry of Health and Family Welfare. (2019). *National Digital Health Blue print*.

3. National Health Authority. (2023). *Ayushman Bharat Health Mission*. Retrieved on 1 September.

Key Enablers of Digital Health in India

India has seen a significant transition in internet usage, increasing from 5% in 2009 to 46% in 2021.⁴ Moreover, there has been a remarkable 200% surge in internet penetration in rural India between 2015 and 2021, underscoring substantial progress towards a digitally connected nation.⁵ In shaping the future of digital health, three pivotal factors - **digital health innovations, Jan Dhan Aadhaar Mobile (JAM) trinity, and increased bank account penetration** - are playing a transformative role, driving accessibility and efficiency in healthcare services.



Rising innovations to enhance digital health



Jan Dhan, Aadhaar, Mobile (JAM) Trinity



Increased Bank Account Penetration

Technologies such as **artificial intelligence, IoT, Blockchain, and cloud computing** are actively enabling the growth of a comprehensive digital healthcare ecosystem. This is resulting in improved access to healthcare services, personalised treatment, early identification of potential diseases and reduced expenses.⁶

IoT is being utilised in India with initiatives like the National Health Stack which aims to create a centralised healthcare system. **Blockchain** is used to secure the health data management system ensuring privacy of the data shared by the patients and **cloud computing** provides accessible and scalable platform for health data storage, and enables cloud-based Telemedicine platforms.

India's robust digital infrastructure that includes **Aadhaar, Unified Payments Interface, and widespread internet/mobile coverage (JAM trinity)**, provides a strong base for the National Digital Health Mission (NDHM). It enables secure authentication, e-signatures, cashless transactions, and efficient healthcare data management.⁶

Launched in August 2014, Pradhan Mantri Jan-Dhan Yojana (PMJDY) aimed to bank unbanked households, forming Digital India's foundation with Aadhaar and mobile connections. Banking services have now been extended to **50.27 Crores beneficiaries** as of 14th September 2023, improving financial access and enhancing healthcare through cashless transactions, expense transparency, and telemedicine.⁷

Key Digital Health Initiatives in India

The introduction of multiple digital initiatives in India has led to the creation of a broader spectrum of digital services within the country, which can be grouped as portals and apps as described below:

Health Portals



National Health portal: A one-stop platform that provides authenticated health information through a single platform for citizens, students, healthcare professionals, and researchers. It covers diseases, health services, tips for healthy living, insurance schemes, and helpline numbers.⁸



Ni-Kshay: A web-based patient management system for Tuberculosis (TB) control that facilitates case registration, test orders, treatment recording, adherence monitoring, and case transfers between care providers. It also acts as the National TB Surveillance System for data reporting to the Government of India.⁹



mDiabetes: A collaborative initiative between the Ministry of Health and Family Welfare (MoHFW) and the WHO, to maintain a portal focused on diabetes prevention and care.¹⁰

4. World Bank. (n.d.). *Individuals using the Internet (% of population) – India*. Retrieved 19 September 2023.

5. India budget. (2023). *Economic Survey 2022-2023*.

6. NITI Aayog. (2020). *National Digital Health Mission-Strategy Overview*.

7. Pradhan Mantri Jan Dhan Yojana. (n.d.). *Pradhan Mantri Jan Dhan Yojana*. Retrieved 14 September 2023.

8. National Portal of India. (2023). *National Health Portal*. Retrieved 15 September 2023.

9. Ni-kshay. (n.d.). *Ni-Kshay-About us*. Retrieved 15 September 2023.

10. National Government services portal. (2023). *mDiabetes*. Retrieved 15 September 2023.

11. Ministry of Health and Family Welfare. (2023). *IDSP Segment of Integrated Health Information Platform*. Retrieved 15 September 2023.

12. Ka, M., Periyasamy et al. (2020). A situation model of integrated health information platform in India: an anticipated review. *International Journal of Community Medicine And Public Health*.

12a Ministry of Health and Family Welfare. (2023). *National Public Health Repository*



Integrated Health Information Platform (IHIP): Previously Integrated Disease Surveillance Programme (IDSP) served as a platform that integrates, coordinates, and correlates information from various sources, both inter-sectoral and intra-sectoral, encompassing public and private healthcare systems in both rural and urban areas. However, challenges identified by the CDC and WHO, such as fragmented data and a lack of epidemiological information, prompted the introduction of IHIP, a web portal developed with WHO that enables real-time surveillance of 36 epidemic-prone diseases using name-based and GIS (Geographic information system)- enabled tracking.^{11,12}



National Public Health Observatory (NPHO): The NPHO and Health Emergency Operation Centre was established for real-time monitoring of data, integrating healthcare programs, and enabling informed decision making to revolutionize healthcare delivery in India.^{12a}



Reproductive and Child Health (RCH) Portal: A web-based portal that promptly recognizes and follow the progress of each beneficiary across the various stages of a woman's reproductive journey. Its primary aim is to encourage, oversee, and provide assistance for the effective execution and reporting of reproductive, maternal, newborn, and child health (RMNCH) programmes.¹³



SUGAM: An e-Governance solution for Central Drugs Standard Control Organisation (CDSCO) that establishes a thorough database for permissions and licenses issued by State FDAs, encompassing information about manufacturers, manufacturing sites, and drug formulations.¹⁴



Health Management Information System: This real-time system optimizes patient treatment flows and empowers the workforce. It benefits recipients and providers, yielding better performance and cost reductions.¹⁵



Intensified Mission Indradhanush (IMI) Portal: A real-time dashboard to enhance nationwide Routine Immunization coverage followed by a **vaccine tracker** for the enhancement of the mission.¹⁶



CoWIN Portal: CoWIN (Covid Vaccine Intelligence Network) is a digital portal for facilitating COVID-19 vaccination. CoWIN portal has been used by 110Cr+ beneficiaries and a total of 220Cr+ vaccination doses have been administered across India using this portal (as of 14th September 2023).¹⁷



eVIN: A system that digitizes vaccine stock management, logistics, and temperature tracking at all levels of vaccine storage. This provides real-time visibility of stock and temperature data, offering an overview of the nationwide vaccine cold chain logistics.¹⁸



National Cold Chain Management Information System (NCCMIS): For tracking the cold chain equipment inventory, availability and functionality.¹⁸

Training and Upskilling Portals



Systematic Assessment of Healthcare Provider's Knowledge and Training (SASHAKT): A training and monitoring web portal to provide e-learning modules for healthcare workers along with real-time updates on the progression of training (launched in the year 2018).¹⁹



Stimulating Advanced Knowledge for Sustainable Health Management (SAKSHAM): An initiative that offers digital health training for healthcare professionals. It operates as a **Learning Management Information System (LMIS)** and serves as a central hub for training materials, case-based learning, best practices, and a database of trained healthcare professionals.²⁰ Currently, SAKSHAM learning management information system hosts more than 200 health-related courses online.²¹



Integrated Government Online Training (iGOT): To address the training requirements of frontline workers, GoI has introduced a training programme for COVID-19 management on DIKSHA (education) platform to enhance the capacity of frontline workers. This programme encompasses a wide range of professionals, including doctors, nurses, paramedics, hygiene workers, technicians, auxiliary nursing midwives (ANMs), state government officers, civil defense personnel, National Cadet Corps (NCC) members, Nehru Yuva Kendra Sangathan (NYKS) participants, National Service Scheme volunteers, Indian Red Cross Society members, Bharat Scouts and Guides, and other dedicated volunteers. The training modules have been designed to offer flexibility, allowing individuals to opt for self-paced learning or on-site training.²²



POSHAN Abhiyaan E-Learning ICMR-NIN: A portal facilitating registered users to access and complete online nutrition e-learning modules hosted by the National Institute of Nutrition (NIN) in Hyderabad.²³



Swayam: Swayam is a significant e-learning initiative by the Indian government, serving as a learning portal for a wide range of online courses. Swayam features several courses related to healthcare, encompassing subjects like public health, health research, dental and healthcare management. Its primary objective is to narrow the digital divide, making education accessible to all students across India.²⁴



Anganwadi Shiksha: Anganwadi workers play a crucial role in looking after the health and nutrition of women and children at community health centers. Unfortunately, they have limited access to organized and consistent training opportunities. To address this deficiency, the government of Madhya Pradesh has introduced an interactive eLearning training platform named "Anganwadi Shiksha" to methodically enhance the capacity of Anganwadi workers.²⁵



Training Management Information System (TMIS): A single platform containing static databases like training guidelines, and course content along with dynamic databases which captures real-time training nominations and evaluations. This serves as a centralised resource to strengthen the public health sector's training and human resources for better service delivery.²⁶



Mobile Academy: A mobile based training programme that aims to update ASHA workers knowledge on life-saving preventive health practices and enhance their interactions with new and expectant mothers and families.²⁷

Major Mobile Apps and Platforms



e-Hospital: Streamlines Indian government sector hospitals with patient-centric care, through a web-based SaaS solution that offers secure data sharing, audit logging of transactions, powerful search facility and tracking of patient history.²⁸



e-Aushadhi: A web-based solution that streamlines the supply chain management of drugs, surgical items, and sutures. It efficiently distributes these supplies to different healthcare centers, ensuring a smooth flow from warehouses to patients.²⁹



Xtended Licensing and Laboratory Node (XLN): A software that is used for monitoring of quality of drugs manufactured or coming from other states and also to ensure reduction in manufacturing of spurious drugs and stop-usage of sub-standard drugs.³⁰



e-Upkaran: A comprehensive software for better inventory management and equipment maintenance in state hospitals. It's a web-based system that streamlines equipment-related processes, benefiting various departments and healthcare centres.³¹



e-RaktKosh: An IT solution designed to standardize and streamline the standard operating procedures, guidelines, and workflow of blood banks nationwide.³²

13. National Health Mission. (n.d.). *Reproductive and Child Health (RCH) Portal*. Retrieved 16 September 2023.

14. Central Drugs Standard Control Organization. (n.d.). *Register Data with State FDA*. retrieved 16 September 2023.

15. Ministry of Health and Family Welfare. (2023). *About Health Management Information System*. Retrieved 16 September 2023.

16. Ministry of Health and Family Welfare. (2020). *Intensified Mission Indradhanush 4.0*. Retrieved 16 September 2023.

17. CoWIN. (2023). *CoWIN dashboard*. Retrieved 14 September 2023.

18. National Health Mission. (2023). *Immunization*. Retrieved 15 September 2023.

19. Ministry of Health and Family Welfare. (n.d.). *Sashakt*. Retrieved 15 September 2023.

20. National Institute of Health and Family Welfare. (2022). *About Saksham*. Retrieved 16 September 2023.

21. Amritmahotsav. (2023). *Launch of SAKSHAM learning management information system*. Retrieved 16 September 2023.

22. DIKSHA. (2023). *iGOT-Health*. Retrieved 16 September 2023.

23. ICMR NIN. (2023). *Poshan Abhiyaan E-Learning ICMR-NIN Modules*. Retrieved 16 September 2023.

24. Swayam. (2023). *Swayam-Course catalog*. Retrieved 16 September 2023.

25. Anganwadi Shiksha-e learning training. United Nations System Standing Committee on Nutrition. (2020). *"Anganwadi Shiksha" e-Learning training to empower front-line nutrition workers in Madhya Pradesh, India*.

26. Ministry of Health and Family Welfare. (2022). *Training Management Information System*. Retrieved 17 September 2023.

27. National Rural Health Mission. (n.d.). *Mobile Academy*. Retrieved 14 September 2023.

28. e-Hospital. (2023). *e-Hospital*. Retrieved 15 September 2023.

29. Central Medical Services Society. (n.d.). *e-Aushadhi*. Retrieved 16 September 2023.

30. e-Gov App Store. (2016). *Xtended Licensing and Laboratory Node (XLN)*. Retrieved 16 September 2023.

31. e-Gov App Store. (2021). *e-Upkaran*. Retrieved 16 September 2023.

32. Ministry of Health and Family Welfare. (2019). *eHealth and Telemedicine*. Retrieved 17 September 2023.

33. e-Gov App Store. (2019). *eCollabDDS*. Retrieved 16 September 2023.

34. Aarogya Setu. (2020). *How Setu Works*. Retrieved 17 September 2023.



eCollabDDS: Web-based platform for real-time collaborative medical and dental image diagnosis. It links remote health centers with tertiary care facilities and enables secondary/tertiary healthcare professionals to collaborate for diagnoses and treatment planning.³³



Aarogya Setu: A mobile app developed by the Indian Government during the COVID-19 pandemic. It offers real-time updates on the pandemic's status, a self-assessment tool, and information about nearby testing centres.³⁴



Mother and Child Tracking System (MCTS): A web-based portal that was developed to deliver comprehensive healthcare and immunization to pregnant women and children up to 5 years old by enhancing and monitoring service delivery while enabling two-way communication between providers and beneficiaries.³²



Miscellaneous Health Apps and Platforms³²

NHP Swasth Bharat App: An app that provides information regarding healthy lifestyle, disease conditions, treatment etc.

NHP Directory services: An app that provides information related to hospitals across India.

Online Registration System: A digital initiative that enables citizens to easily access various services online, such as appointment booking, fee payments, viewing diagnostic reports, and checking blood availability at public hospitals.

No more Tension mobile app: App for stress management.

Pradhan Mantri Surakshit Matritva Abhiyan Mobile App: Voluntary engagement app for PMSMA Programme.

Kilkari: A health education mobile app used for delivering timely, accurate information on reproductive, maternal, neonatal, and child health.

ANM Online application-ANMOL: A tablet app that enables Auxiliary Nurses and Midwives (ANM) to directly enter and update beneficiary data in their jurisdiction.

India Fights Dengue App: App for empowering community against dengue.

My (Mera) hospital (Aspataal): A Government of India initiative that collects patient feedback on hospital services through SMS, mobile apps, and a web portal. Patients can provide feedback in seven languages for hospitals visited in the last seven days.

mCessation: Mobile-based strategy known that aims to motivate and assist individuals who wish to quit smoking

eSushrut, Hospital Management Information System: Used for integrating computerized clinical information system for improved administration and patient care.

NCD Application: An application used by Ayushman Bharat Comprehensive Primary Healthcare (CPHC) programme to create electronic health records for individuals over 30. This technology allows for efficient data collection during screenings and treatments, aiding health officials in measuring disease burden, programme planning, and evaluating health worker performance.

Poshan Tracker: Poshan Tracker is a mobile app launched by the Ministry of Women and Child Development, Government of India, on March 1, 2021. It's a crucial tool for dynamic identification of child malnutrition, underweight and monitoring nutrition service delivery at the last mile.

MusQan: The MusQan initiative, established by the Government of India under the National Health Mission (NHM), focuses on improving the quality of child healthcare at public health facilities. It employs various assessment tools and aims to provide high-quality, child-friendly services, reduce child morbidity and mortality, and strengthen healthcare standards and infrastructure.

NOTTO platform: A National Organ and Tissue Transplant Organization's web platform that encourages organ donation through online services, including registration organ/tissue transplantation or retrieval and pledge registry.

FP-LMIS (Family Planning- Logistics Management Information System) app: An application to manage supply chain operations and regulate the flow of family planning supplies from national level to end users.³⁵

Central Dashboard: Developed to monitor key health programme indicators by collecting data from existing public health information systems. It also aids in analyzing the progress of new healthcare initiatives introduced by the Ministry of Health and Family Welfare, Government of India.

35. National Health Mission. (2017). *Operational Guidelines for Rolling out Family Planning Logistics Management Information System*.

Trends in Digital Health

Digital technology is emerging as a game-changer in healthcare delivery in India, and at the forefront of this transformation is Ayushman Bharat Digital Mission (ABDM), serving as the backbone of the country's digital health advancements by seamlessly integrating unique IDs, registry systems, and data analytics. Paired with telemedicine, these efforts are collectively enhancing access to these ground-breaking features of modern healthcare.



1. Introduction of Ayushman Bharat Digital Mission

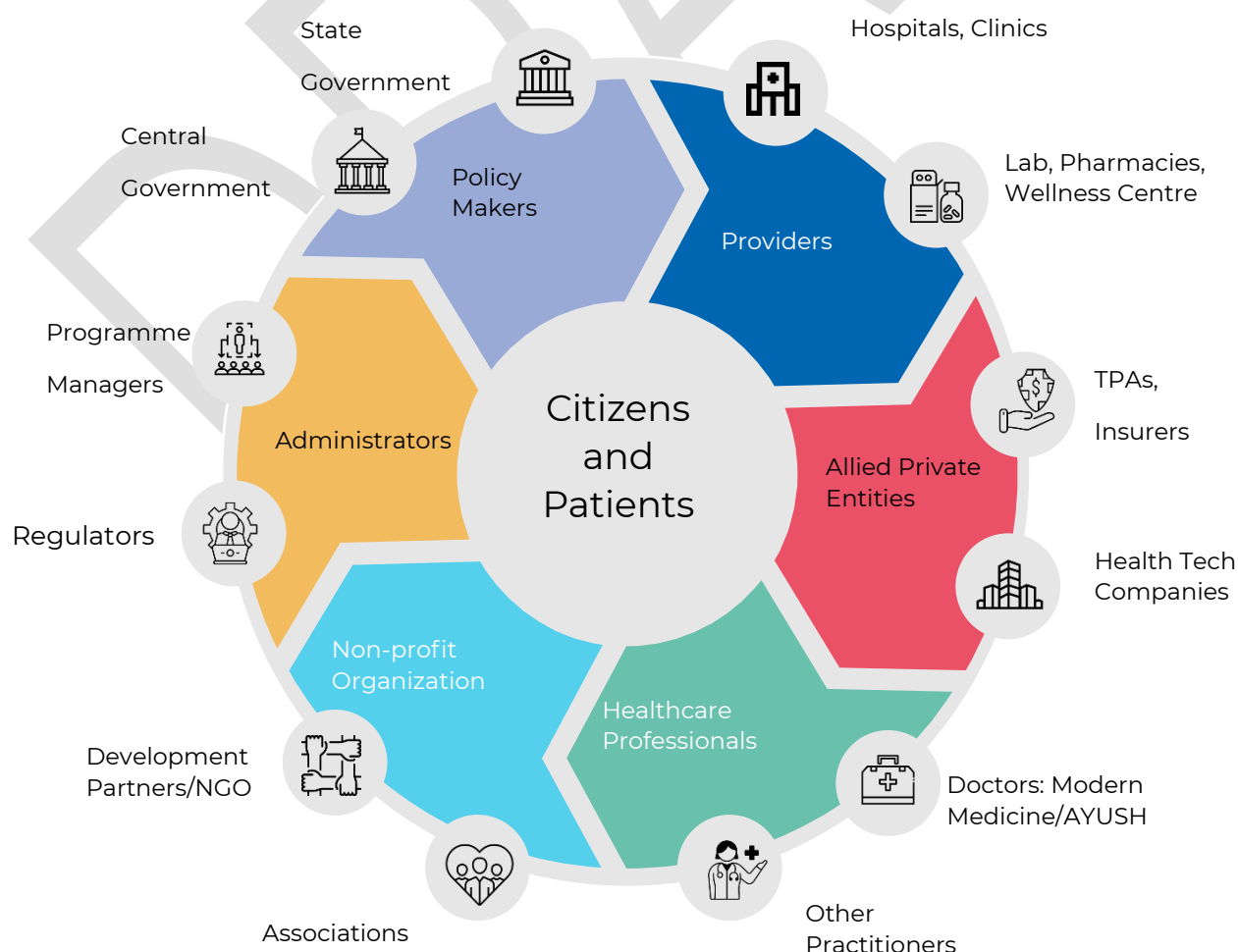
The Ayushman Bharat Digital Mission (ABDM) is a national digital health ecosystem that was launched on March 2020, which connects the digital health solutions of healthcare providers across the country and provides a wide range of data, information, and infrastructure services to support universal health coverage

Building Blocks

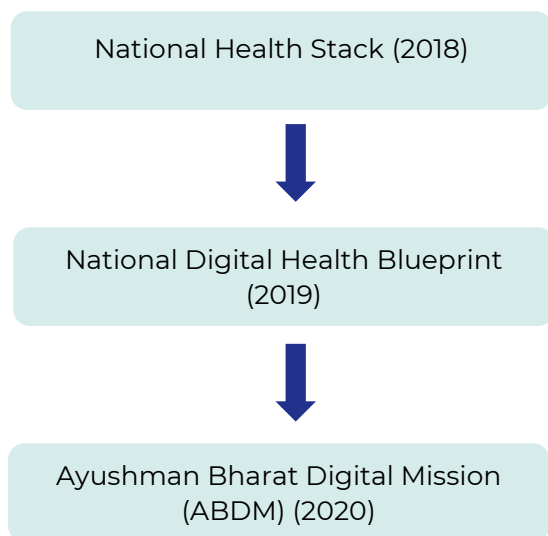
- **ABHA Number:** Patients are allotted Ayushman Bharat Health Account (ABHA) number to uniquely identify medical records and ensure that the created medical records are issued to the correct individual.³
- **Healthcare Professionals Registry (HPR):** A comprehensive repository of all healthcare professionals involved in delivery of healthcare services. This registry will enable them to get connected to India's digital health ecosystem.³
- **ABHA Mobile App (PHR):** An electronic record of health-related information on an individual that conforms to nationally recognized interoperability standards and that can be drawn from multiple sources while being managed, shared, and controlled by the individual.³
- **Health Facility Registry (HFR):** A comprehensive repository of health facilities of the nation across different systems of medicine (both private and public facilities).³



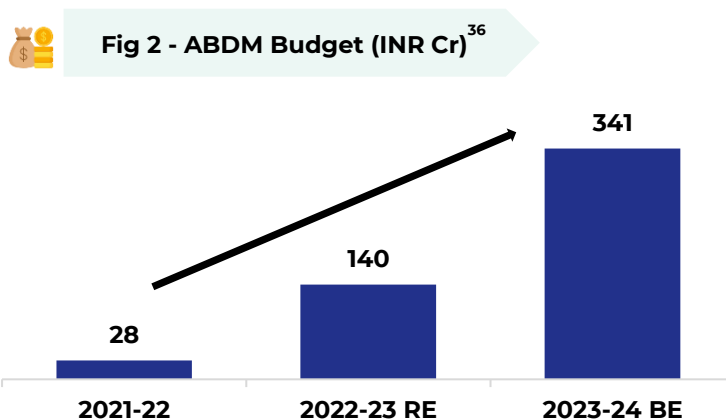
Fig 1 - Ayushman Bharat Digital Mission (ABDM) Ecosystem³



Strategic Evolution of ABDM



RE: Revised Estimate, BE : Budget Estimate



- ABDM was rolled out nationally in 2022 under the National Health Authority with an allocation of **INR 1,600 Crores** for 5 years³⁷
- The proposed allocation for ABDM has increased by 144% from revised estimates of 2022-23 to INR 341 Crores in 2023-24

Ayushman Bharat Digital Mission (ABDM) has successfully integrated 125 digital health service applications and there are a total of 1093 Active Integrators in the ABDM Sandbox(as of 19th September, 2023). These include both Public and private application.³⁸

2. Adoption of Telehealth /Telemedicine

Telemedicine in India has gained significant traction and is experiencing a notable rise in popularity and adoption. Major hospitals like Apollo, AIIMS, Narayana Hrudayalaya have adopted telemedicine services and entered into a number of Private-Public Partnerships³⁹

Major Government Initiatives



eSanjeevani: The world's largest recorded telemedicine implementation functioned through two models: first, a patient-to-doctor telemedicine approach (eSanjeevani OPD), and second, an assisted telemedicine (eSanjeevani AB-HWC) focused on delivering primary healthcare services. This has reached the landmark milestone of providing telemedicine services to more than 10 Crores patients.⁴⁰

eSanjeevani AB-HWC consultations: 9,04,18,022 and eSanjeevani OPD consultations: 96,99,653 (as of Feb'23)⁴⁰



SeHAT: e-Health Assistance and Tele-consultation Service is a tri-services teleconsultation service for the Armed Forces of India. Presently, it is designed to fulfil the needs of general OPD consultation. However, there are plans to expand it to include specialist OPD consultation in the future.

Launched in May 2021⁴¹

36. India Budget. (n.d.). Ministry of Health and Family Welfare- Demand no. 46: Department of Health and Family Welfare 2023-2024.

37. Invest India. (2023). India's Healthcare Industry: Navigating the Road to 'Health for All'. Retrieved on 5th September 2023.

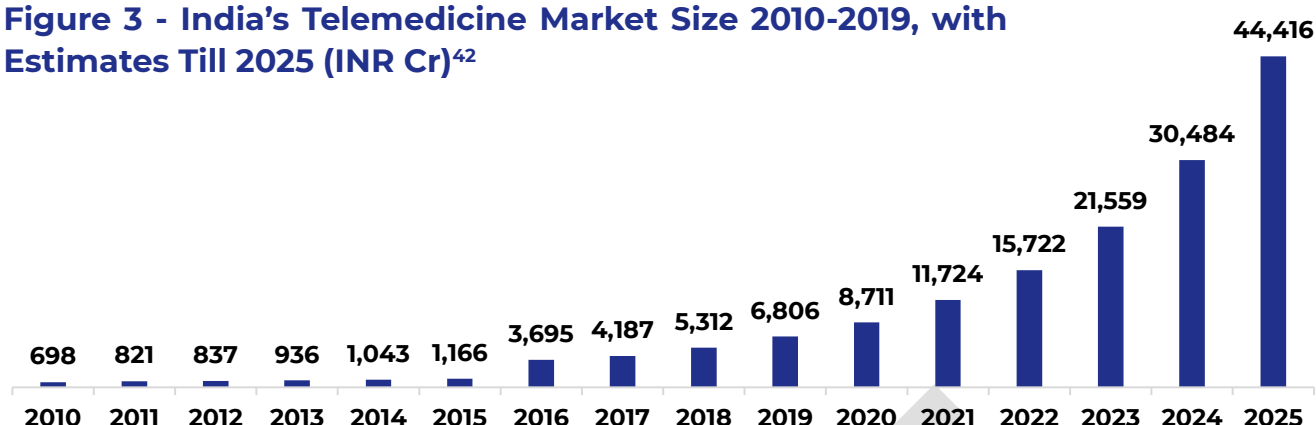
38. Ayushman Bharat Digital Mission. (n.d.). ABDM Sandbox Dashboard. Retrieved 17 September 2023.

39. Shetty KP (2022). A Study on the Transformation of the Indian Healthcare Sector during the COVID-19 Pandemic. International Journal for Research in Applied Science and Engineering Technology

40. Centre for Development of Advanced Computing. (2023). Sanjeevani Landmark Milestone. Retrieved 17 September 2023.

41. Sehat. (2021). SeHAT. Retrieved 17 September 2023.

Figure 3 - India's Telemedicine Market Size 2010-2019, with Estimates Till 2025 (INR Cr)⁴²



Future Growth

- Telemedicine market in India is expected to reach a valuation of INR 44,416 Crores by 2025, growing at a CAGR of 31%.
- Telemedicine initiatives are being undertaken by several private sector companies such as Practo, Tata 1mg, and Apollo 24x7. Among these, Pharmeasys has achieved a valuation of INR 1,231 Crores in the field of telemedicine.⁴³

eSanjeevani- National Telemedicine Service

- A national telemedicine service that promotes digital health equity and Universal Health Coverage (UHC) and offers convenient access to doctors and specialists via smartphones and facilitates remote healthcare through Ayushman Bharat Health and Wellness Centres (HWCs).
- As of February 2023, **1,15,234 Health and Wellness Centres** (acting as spokes) served 100.11 million patients. These centres were connected through 15,731 hubs, along with 1,152 online OPDs staffed by 2,29,057 medical specialists and super-specialists trained in telemedicine.⁴⁰

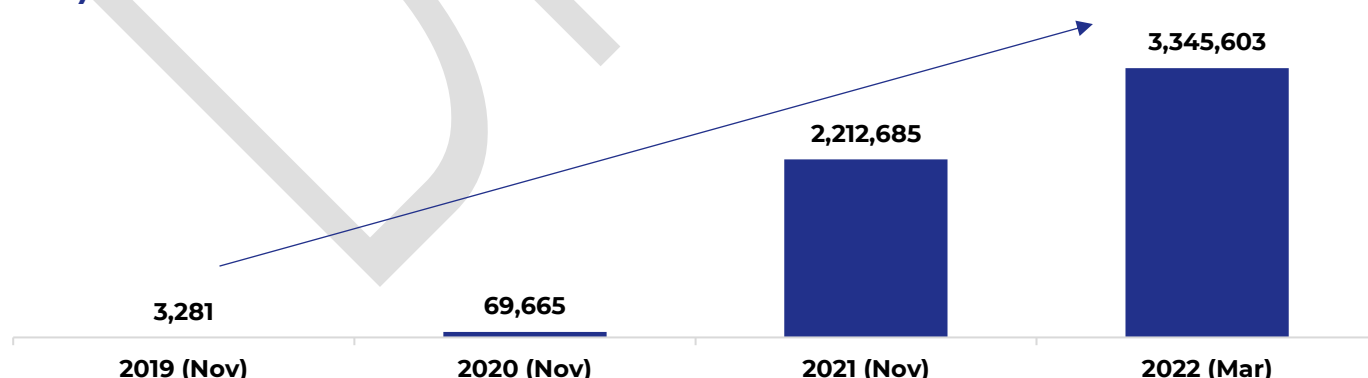
1,07,725 Spokes
Operationalized⁴⁴

1,91,450 Providers
Onboarded⁴⁴

124 Specialties⁴⁴

Data updated: 19th September 2023

Figure 4 - Teleconsultation Conducted Through eSanjeevani (Updated till March 2022)⁴⁵



Telemedicine Guidelines

In March 2020, the Government of India has published Telemedicine Practice Guidelines in supersession of the Medical Council of India.⁴⁶

42. NITI Aayog. (2021). *Investment Opportunities in The Healthcare Sector*.

43. InvestIndia. (n.d.). *The Indian Unicorn Landscape*. Retrieved 8 September 2023.

44. Ministry of Health and Family Welfare. (2022). *eSanjeevani*. Retrieved 19 September 2023.

45. National Health Mission. (2022). *Ayushman Bharat Health and Wellness Centres*.

46. Ministry of Health and Family Welfare. (2020). *Telemedicine Practice Guidelines*.

Applications of Telehealth / Telemedicine

Telemedicine technology has access to various treatment options, including primary care consultations, psychotherapy, physical therapy through wireless technologies and below mentioned are some of the major applications in telemedicine:

Tele Mental Health⁴⁷

- Union Ministry of Health and Family Welfare is providing Tele Mental Health Assistance and Networking Across States (Tele-MANAS) helpline and has reached a milestone by receiving over 269039+ calls since its launch in October 2022.
- Presently there are 51 functional Tele MANAS Cells across 27 States and UTs, providing mental health services in over 20 languages and over 1,600 trained counsellors running first-line services.

e-Aushadhi Drugs and Vaccines Distribution and Management System

- The provision of free generic drugs through public health facilities is one of the most cost-effective ways to reduce Out Of Pocket Expenditure (OOPE), particularly for poor and vulnerable groups.
- DVDMS is a web-based supply chain management system developed by CDAC that deals with the purchase, inventory management and distribution of various drugs, sutures, and surgical items across all government health facilities and Drug Distribution Counters (DDCs) within the State/UT.
- MoHFW approved the roll-out of the “National Health Mission – Free Drugs Service Initiative (FDSI)” under which the States / UTs are being supported to provide free generic drugs to patients coming to public health facilities.
- However, there has been a rise in selling of medicines from unlicensed premise. Multiplicity of dispensing of pharmacies, misuse of drugs and increase in drug addicts for which DCC organized a sub-committee for regulation of e-pharmacies in India.⁴⁸

Telepathology

- A recent addition to pathologist's diagnostic tools that helps in building bridge between the rural and urbanized sections in seeking quality diagnostic services.
- It allows for emergency pathological analyses, like intraoperative frozen section, to be performed without the presence of a pathologist, addressing the lack of primary medical care in rural areas.⁴⁹

Teleradiology

- To meet the imbalance between the demand and availability of diagnostic services, teleradiology was brought into place.
- The Ministry of Health and Family Welfare has launched free Tele-Radiology services under the Free Diagnostics Service Initiative programme, currently available in 14 states / union territories across India.⁵⁰

Remote home medical care

- The growth of the elderly population and the shortage of specialist physicians make home care through TLM a promising field.
- This type of treatment is essential for most patients with cardiovascular disease and chronic respiratory diseases.⁵¹

Remote Patient monitoring

- Telemedicine (TLM) can be used to remotely monitor patients of all ages and conditions, especially in remote areas where there is a lack of specialised doctors.
- In case of medical emergencies, patients can be examined by experienced physicians assigned at medical centres with this level of service. Through this facility, doctors can also provide first level of healthcare treatment to patients living at distant places.⁵¹

47. Ministry of Health and Family Welfare. (2023). *Tele-MANAS-dashboard*. Retrieved 8 September 2023.

48. Central Drugs Standard Control Organization. (2018). *Report of the Sub Committee constituted by the Drugs Consultative Committee to examine the issue of regulating the sale of drugs over internet under the drugs and cosmetics rules, 1945*.

49. Feroz et al. (2020). Telepathology: An update on applications, latest advances, and current status in Indian scenario. *Journal of cancer research and therapeutics*, 16(4), 703–707.

50. National Health Systems Resource Centre. (2023). *Free Diagnostic Service Initiative*. Retrieved 9 September 2023.

51. Jafarzadeh et. al. (2022). Different applications of telemedicine - assessing the challenges, barriers, and opportunities- a narrative review. *Journal of family medicine and primary care*, 11(3), 879–886.

- To ensure timely healthcare services for pregnant women, lactating mothers, and children. MCTS, a web-based tracking system, was introduced nationwide to provide real-time information on services and vaccinations delivered to pregnant women and children.
- An upgraded version, the RCH Portal, was launched by the Ministry to enable early identification and continuous tracking of beneficiaries throughout their reproductive lifecycle.
- The MCTS/RCH portal collects beneficiary data, which is utilized by various applications like ABDM (Ayushman Bharat Digital Mission), eJanma (Karnataka), Kutumba (PDS system of Karnataka), UMANG, ANMOL, MCTFC, Kilkari, and Mobile Academy. The RCH Portal data is pivotal for the direct transfer of Janani Suraksha Yojana (JSY) benefits to pregnant women and payments to ASHAs, wherever applicable.
- This portal has been successfully implemented in 30 States/Union Territories, with Rajasthan and Gujarat transmitting their data through web services. Additionally, four states, namely Madhya Pradesh, West Bengal, Tamil Nadu, and Telangana, are in the integration process with the RCH Portal for data transmission.
- Health workers utilize the portal to plan service delivery and identify beneficiaries due for Antenatal check-ups, postnatal check-ups, immunization services and also aids in identifying high-risk pregnant women, monitoring health conditions.

ANM On Line (ANMOL)⁵²

- A tablet-based application designed to enhance the efficiency and effectiveness of Auxiliary Nurse Midwives (ANMs) who are frontline healthcare workers responsible for various daily tasks. This application enables ANMs to enter and update beneficiary service records in real or near-real-time.
- In addition to record-keeping, ANMOL serves as a valuable job aid for ANMs by providing them with readily accessible information such as due lists, dashboards, and guidance based on the data they input. This standardizes the provision of maternal and child care services by ANMs, allowing them to plan Village Health and Nutrition Days (VHND) and manage vaccines and logistics effectively.
- It also offers audio and video counselling features, which play a crucial role in raising awareness among beneficiaries about various government schemes.
- ANMOL is currently operational in 30 States/Union Territories and is available in three different languages: English, Hindi, and Kannada.

Kilkari and Mobile Academy⁵²

- **Kilkari**, translating to "the joyful sound of a baby," provides families with complimentary weekly audio messages through mobile phones about pregnancy, childbirth, and child care, tailored to the specific stage of pregnancy (starting from second trimester of pregnancy and continuing till one year of age).
- Both Kilkari and Mobile Academy, were officially launched by the union Health minister on January 15th, 2016.
- Collectively, Kilkari and Mobile Academy are enhancing family well-being across various aspects such as family planning, reproductive health, maternal and child health, nutrition, and sanitation.
- Kilkari has been implemented in 18 States/Union Territories, aiming to disseminate critical health information and support.
- **Mobile Academy** offers a valuable audio training course that is specifically designed for Accredited Social Health Activists (ASHAs) to enhance and update their knowledge base and communication skills by making this training opportunity accessible via their mobile phones.
- Mobile academy operates in 17 States/Union Territories, concurrently benefiting numerous ASHAs by providing training via mobile devices.

52. Ministry of Health and Family Welfare. (2023). *MMPC*.

Opportunities and Challenges

Opportunities^{55,59}



Reduced Cost

Introduction of digital platforms across different areas can enable remote consultations, allowing patients to access healthcare professionals without the need for in-person visits. This can reduce travel costs and the need for physical infrastructure, resulting in cost savings for both patients and healthcare providers.



Increased Awareness

Digital platforms provide opportunities to disseminate health education and awareness programmes at scale. Mobile applications, websites, and social media platforms can be utilized to educate individuals about preventive care, healthy lifestyles, and make them aware about disease management.



Better Access to Quality Care

Digital health solutions like EHR, helps to promote a seamless exchange of patient information between healthcare providers, ensuring continuity of care and reducing medication errors. This can reduce the need for paper-based documentation and enable quick access to patient information, leading to more efficient healthcare delivery.



Seamless Transaction

Digital payment solutions offer significant help to patients by allowing them to make payments easily through various digital channels such as mobile apps, online portals, or contactless payment that helps to streamline the payment process, enhance transparency and enable patients to track their expenses and access billing information conveniently.

53. Ministry of Health and Family Welfare. (2018-2019). *E-Governance and Telemedicine*.

54. Ministry of Health and Family Welfare. (n.d.). *Dell EMC- CPHC NCD Solution*.

55. Ministry of Health and Family Welfare. (2017). *National Health Mission Annual Report 2017-2018*.

56. Ministry of Ayush. (2021). *Framework for Implementation and Operational Guidelines Ayush Oushadhi Guvvata Evam Utpadan Samvardhan Yojana*.

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59. G20. (2023). *G20 Leaders Declaration*.

58. Digital Health Atlas. (n.d.) *Digital Health Atlas*. Retrieved 17 September 2023.

Challenges



Digital Divide

The digital divide, stemming from various socio-economic reasons like: inadequate literacy, digital skills, limited internet coverage, financial instability and scarce mobile phone access, can impede both citizens and providers.



Language and Literacy

India is a linguistically diverse country with several regional languages. Creating digital health platforms that are accessible in multiple languages is essential to cater to the linguistic diversity of the population. Additionally, the literacy levels vary, and designing user-friendly interfaces and tools that can be easily understood and navigated by individuals with varying levels of literacy is crucial.



Data Security and Privacy

Digital health involves the collection, storage, and sharing of sensitive health data. Ensuring robust data security measures and protecting patient privacy is critical to gain trust of individuals and healthcare providers. While the government has recently enacted the Digital Personal Data Protection (DPDP) Act⁶² to safeguard personal data, there is still a necessity for more such legislations and public awareness to ensure solution providers protect patient data and prevent misuse.



Maintenance and Sustainability

Digital health solutions require ongoing maintenance like regularly updating software, applying security patches, backup and disaster recovery, system performance monitoring etc. to ensure their continuous functioning and effectiveness. Developing sustainable business models in the health domain, addressing infrastructure challenges, and securing long-term funding are essential for the successful implementation and sustainability of digital health initiatives.

Lack of Infrastructure

Ayushman Bharat Digital Mission implementation is hindered by lack of software and hardware infrastructure in hospitals compounded by connectivity issues in remote areas. There is a lack of adoption of Hospital Management Information System (HMIS) in the hospitals.

Resistance to Change

Resistance to adopt ABDM stems from the significant behavioral change which would be required to successfully adopt ANDM, which requires the doctors to use digital devices for prescriptions. Incentivising this change, particularly amidst the heavy workload, poses a considerable challenge for doctors, nurses, and pharmacy staff.

59. G20. (2023). *G20 Leaders Declaration*.

60. Apollo. (2021). *Apollo Public Private Partnership*. Retrieved 19 September 2023

61. Eka care. (2023). *Eka Care Services*. Retrieved 19 September 2023

62. Ministry of Electronics and Information Technology. (2023). *Digital Personal Data Protection Act*.

63. Ministry of Electronics and Information Technology. (n.d.). *India's Trillion Dollar Digital Opportunity*.

Cross-Cutting Themes

Policy and Regulation



Significance: Establishing robust policies and regulatory frameworks to govern digital health practices.

Current Efforts: National Digital Health Mission, Telemedicine, E-governance are some of the initiatives towards digital health in India.

Way Forward: Refining policies to keep pace with technological advancements and safeguard patient interests.

Education and Awareness



Significance: Digital tools help strengthen health literacy and accessibility, empowering individuals to make informed health decisions. It is also beneficial for educating health professionals.

Current Efforts: Saksham is a dedicated and unified platform for providing online training and medical education to all health professionals in the country.

Way Forward: Expanding digital health education into remote areas. Customising the content according to demography and language will accelerate the adoption of healthy practices.

Research and Development



Significance: Digital platforms aid research innovation and improved healthcare solutions.

Current Efforts: Research and development for multiple initiatives like Telemedicine, Ayushman Bharat Digital Mission, etc.

Way Forward: Encouraging public-private partnership can encourage tech-driven R&D.

Data Security and Privacy



Significance: Safeguarding patient's data and ensuring privacy in digital health systems.

Current Efforts: National Digital Health Blueprint emphasises the significance of data security and is working on making the protection standards mandatory.

Way Forward: Strengthening data security frameworks and promoting compliance with global data protection standards.

Ministries and Departments

Convergence among Indian ministries and departments for policy making is an important aspect of governance in India. It involves the coordination and collaboration between different ministries and departments to achieve common goals and objectives.



Ministry of Health and Family Welfare

Department of Health and Family Welfare

- Ensuring the availability and accessibility of quality healthcare services to all citizens through government hospitals, health centers, and other healthcare facilities.
- It mainly functions in implementing health policies, family planning programmes, disease surveillance, Health Promotion and Education etc.

Department of Health Research

- Responsible for promoting and coordinating health research activities in the country.
- Supports and strengthens research in various areas of health and medicine to address public health challenges and improve healthcare services.



Ministry of Electronics and Information Technology

Department of Electronics and Information Technology

- Responsible for formulating and implementing policies, programs, and initiatives related to electronics, information technology, and IT-enabled services.
- Plays a vital role in driving India's digital transformation, promoting domestic electronics manufacturing, enhancing cybersecurity, fostering innovation and start-ups, and ensuring digital literacy for all citizens.



Ministry of Communications

Department of Telecommunications

- Structures and implements policies for telecommunications and information technology.
- Manages and improves telecommunication infrastructure, promoting innovation, and fostering investment and competition in the sector.



Ministry of Defence

Ministry of Defence

- Formulates defense policies, manages armed forces, and ensures readiness. Its main goal is to safeguard the nation and maintain peace.
- Oversees defense procurement, budget allocation, and technology upgradation to keep the armed forces equipped with modern weaponry and equipment.



Prime Minister's Office

Department of Space

- Implements policies related to space research, exploration, and applications to drive technological advancements and address the country's developmental challenges.
- Oversees ISRO and focuses on using space technology for national development and societal benefits. It has achieved significant milestones and contributes to various sectors, including communication, agriculture, and weather forecasting.



Ministry of Chemicals and Fertilizers

Department of Pharmaceuticals

- Focuses on promoting the availability of essential and affordable medicines to the people and ensuring the growth and development of the pharmaceutical sector.
- Collaborates with various stakeholders, including pharmaceutical companies, research institutions, and regulatory bodies, to facilitate the production of high-quality medicines and promote research and innovation in the pharmaceutical field.

Government Interventions



Government Initiatives on Telemedicine

- **National Medical College Network (NMCN):** Established to provide e-Education and e-Healthcare delivery, wherein 50 Government Medical Colleges are being interconnected, over NKN (National Knowledge Network) having high-speed bandwidth connectivity.⁵³
- **National Telemedicine Network (NTN):** An initiative approved with the vision to provide telemedicine services to remote areas by upgrading existing government healthcare facilities.⁵³
- **CORS (CollabDDS Online Radiology Services):** A web interface among different health communities for resolution of radiological and dental issues making it accessible to local as well as remotely situated doctors for seeking guidance from expert radiologists.⁵³
- **SATCOM Based Telemedicine Nodes:** An initiative to build telemedicine nodes at pilgrimage places using space technology tools to bridge a gap between identified remote patient and health facilities and specialty hospitals in collaboration with the Department of Space for health awareness.⁵³
- **JIPMER BIMSTEC:** MoHFW inaugurated the JIPMER-BIMSTEC Telemedicine Network to improve regional cooperation in the field of healthcare by strengthening telemedicine-based patient care services.⁵³
- **TeleEVIDENCE:** A platform via which doctors can testify in the judicial process utilizing video conferencing facility without visiting the courts⁵³
- **Some other Teleconsultation Platforms:** eRaktkosh (manages all the blood banks across the country), ORS (Online Reservation System application that provides for online appointment for government facilities across the country), Mera Aspataal (platform for giving feedback on health services provided by the hospitals).⁵³



Centre for Health Informatics Initiatives

- **Health and Wellness Centers (HWC):** Initiated a portal/website to monitor the progress of Health and Wellness Centers (HWCs) under Ayushman Bharat Scheme.⁵³
- **CPHC-NCD Programme:** A programme to provide necessary software, and third-party software for prevention, control and screening for Non-Communicable Diseases (NCD), developed by DELL for Ministry of Health and Family Welfare.⁵⁴
- **LaQshya:** An initiative to develop dashboard to improve the quality of care in labour room and maternity operation theatres.⁵³
- **National Programme for Health Care of Elderly (NPHCE):** Portal for monitoring the Electronic Health Record (EHR) Data of elderly.⁵³

SATCOM : satellite communication , BIMSTEC : Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation , CPHC : Comprehensive Primary Healthcare, DDS : Digital Diagnosis System , AEFI : Adverse Events Following Immunization

53. Ministry of Health and Family Welfare. (2018-2019). *E-Governance and Telemedicine*.

54. Ministry of Health and Family Welfare. (n.d.). *Dell EMC- CPHC NCD Solution*.

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58. Digital Health Atlas. (n.d.). *Digital Health Atlas*. Retrieved 17 September 2023.

59. G20. (2023). *G20 Leaders Declaration*.



Emerging Digital Initiatives

- With respect to supply chain management, the government has introduced the **web-based DVDMS system**, optimizing the procurement, inventory, and distribution of medical essentials. Furthermore, the "e-Aushadhi" initiative enhances healthcare management effectiveness.^{55,56}
- **EMR website** and dashboard is still under development.⁵³
- Developed **VAEIMS (Vaccine Adverse Event Information Management System)** to facilitate the collection, collation, transmission, analysis and feedback of vaccine safe data (AEFI data).⁵³
- NITI Aayog is currently at an advanced stage in its efforts to launch a programme aimed at establishing a comprehensive **national repository of annotated and curated pathology images**. Additionally, there are ongoing discussions regarding another project, namely, the creation of an **Imaging Biobank for cancer**.⁵⁷

Digital Health infrastructure is also driven by various global initiatives and Public-Private Partnerships



Prominent Global Digital Health Initiatives



Global Digital Health Partnership (GDHP) is a collaboration of governments and organization across the globe to support the executive implementation of worldwide digital health services.⁵³

Digital Health Atlas (DHA) is an open-sourced platform for global digital health coordination.⁵⁸

The **Global Initiative on Digital Health (GIDH)** is set up under WHO to create a full digital health ecosystem while adhering to data protection regulations in India.⁵⁹



Indian Public – Private Partnerships



Apollo TeleHealth has partnered with central and state governments for projects like digital dispensaries, Tele-medicine, etc.⁶⁰



Eka Care became the first CoWIN-approved organization in India, through which users could book their vaccination slot, download their certificate, and even create their Health IDs.⁶¹

AI Based Health Care

Utilisation of AI is now becoming the new trend in healthcare, leading to more personalised treatment, early identification of potential pandemics, and advanced imaging diagnostics making it accessible to even in rural areas. The synergy of AI, robotics, and the Internet of Medical Things (IoMT) could potentially become the new revolution in healthcare by providing innovative solutions to address healthcare challenges and supporting the government in achieving its objectives.⁵⁷

Safety Assurance

Development of robust laws and policies for the secure and safe collection, storage, and management of personal data has become a very important component to instill confidence in digital platforms and protect user privacy. According to NHDB, ensuring safety in healthcare services and electrical-medical equipment is significant. Making the equipment's used comply with safety standards, and incorporating Standard Treatment Guidelines into clinical workforce can ensure standardized, safe patient care, instilling confidence in both patients and care providers.^{2,63}

Real Time Monitoring

Nowadays real-time data visualization and deep analytics capabilities are being harnessed to track and monitor national projects effectively.⁶³

Infrastructure

Digitally enabled power access is becoming the new way for improving the financial condition of distribution companies in India, leading to improvement in infrastructure for power access.⁶³

Integrated Platform

Introduction of an integrated digital education platform with customizable local-language content is now an open societal platform for healthcare teachers, learners, and healthcare providers.⁶³

Wireless Connectivity

A combination of fibre to the home (FTTH) and wireless technology are most recent advancements that are being used in last-mile telecommunications. With this technology in hand, it becomes easy and affordable for people to access digital health even in rural areas.⁶³

60. Apollo. (2021). *Apollo Public Private Partnership*. Retrieved 19 September 2023

61. Eka care. (2023). *Eka Care Services*. Retrieved 19 September 2023

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Best Practices

1. Core Dash Board-Integrated Hospital Sanitation Monitoring System in Andhra Pradesh^{64,65}

Background

Maintaining cleanliness and hygiene in healthcare facilities plays a crucial role in infection control, preventing the spread of diseases within the healthcare facility, thus helping patients to recover more effectively in clean environments, and instil confidence in the healthcare system.

Steps taken by the State Government

- To reduce Hospital Acquired (Nosocomial) infections, the government of Andhra Pradesh has developed and endorsed the New Scientific Sanitation Policy-2015 to enhance sanitation practices in government hospitals.
- Alongside, the government also developed a web-based Hospital Sanitation Monitoring System dashboard to keep a real-time track of sanitation, security and pest and rodent control services which will be implemented by different service holders to create a visible impact in providing satisfaction to patients.
- This real-time monitoring tool have been grouped within the ten parameters according to the weightage. Each monitoring items further has specific measurable elements. These measurable elements are checked in each department of a health facility through specific checkpoints. All Checkpoints for monitoring items are collated, and together they form assessment tool called 'Checklist'. Scored/ filled-in Checklists would generate scores reflecting percentage of performance along with hospital average score (Sanitation, Security and Rodent control).
- The scores entered by facilities will be then reflected on the core dashboard along with facility photographs.

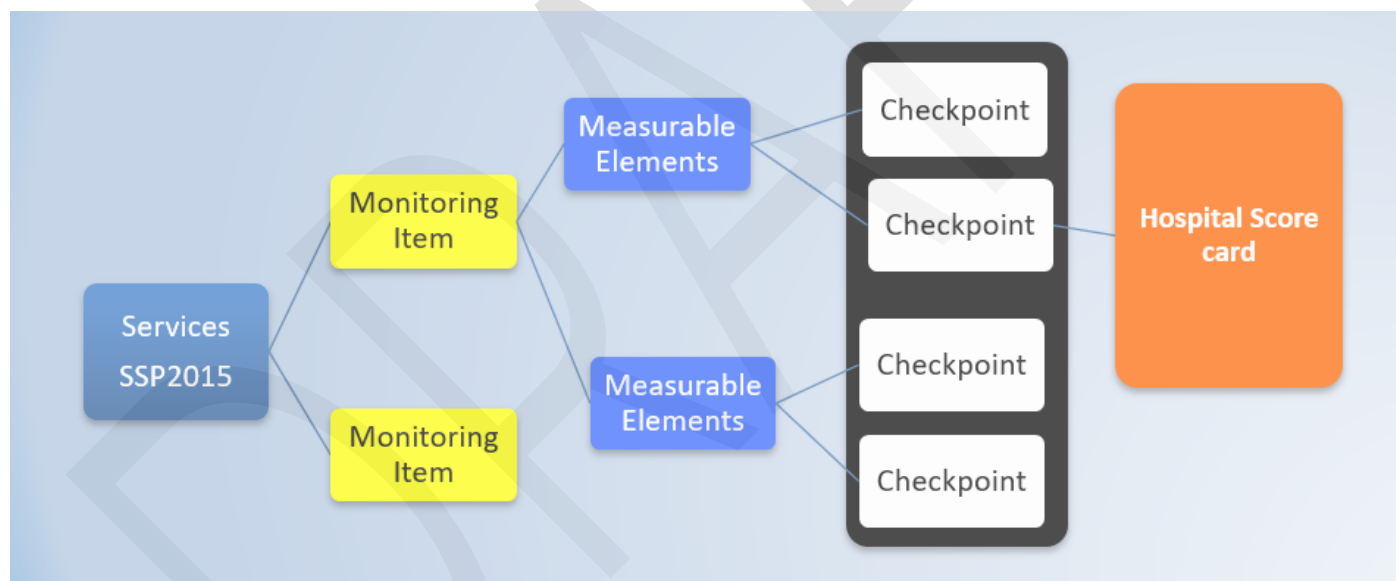


Figure 5 - Image showing the workflow for the Hospital Sanitation Monitoring System

Impact generated

Implementation of this dashboard brought potential changes in the environment of public healthcare facilities:

- It has elevated public perception and patient satisfaction concerning sanitation and security services at public healthcare facilities.
- A significant reduction in hospital Acquired Infections(HAI) were also observed in the hospitals where HSMS was implemented.

64. National Health System Resource Centre. (2021). *National Summits on Good Replicable Practices & innovations in Public Healthcare Systems in India*.
65. DMEQ. (2021). *Integrated Hospital Sanitation Monitoring System Dashboard*.

2. Dakshata Mentor Application- Rajasthan⁶⁶

Background

Despite significant advancements in maternal and neonatal outcomes in India over the last decade, the current maternal mortality ratio and neonatal mortality rate still fall short of the Sustainable Development Goal targets. This discrepancy can be attributed to less optimal quality of maternity care. Despite progress in several areas, challenges such as limited access to quality healthcare facilities, insufficient training of healthcare professionals, and socioeconomic disparities have hindered India's ability to meet these crucial health-related goals.

Steps taken by the State Government

- To bring a reduction in maternal and newborn mortality, the Government of India (GoI) has introduced a strategic initiative known as 'Dakshata' that aims to empower healthcare workers in facilities to deliver high-quality care during childbirth.
- Its main goal is to strengthen facility-level mentoring and supervision of labour room through user friendly application which allows efficient data collection and generates actionable outputs. This application acts as a consistent mentoring system that ensures delivery of high-quality intra-natal and immediate postpartum care.

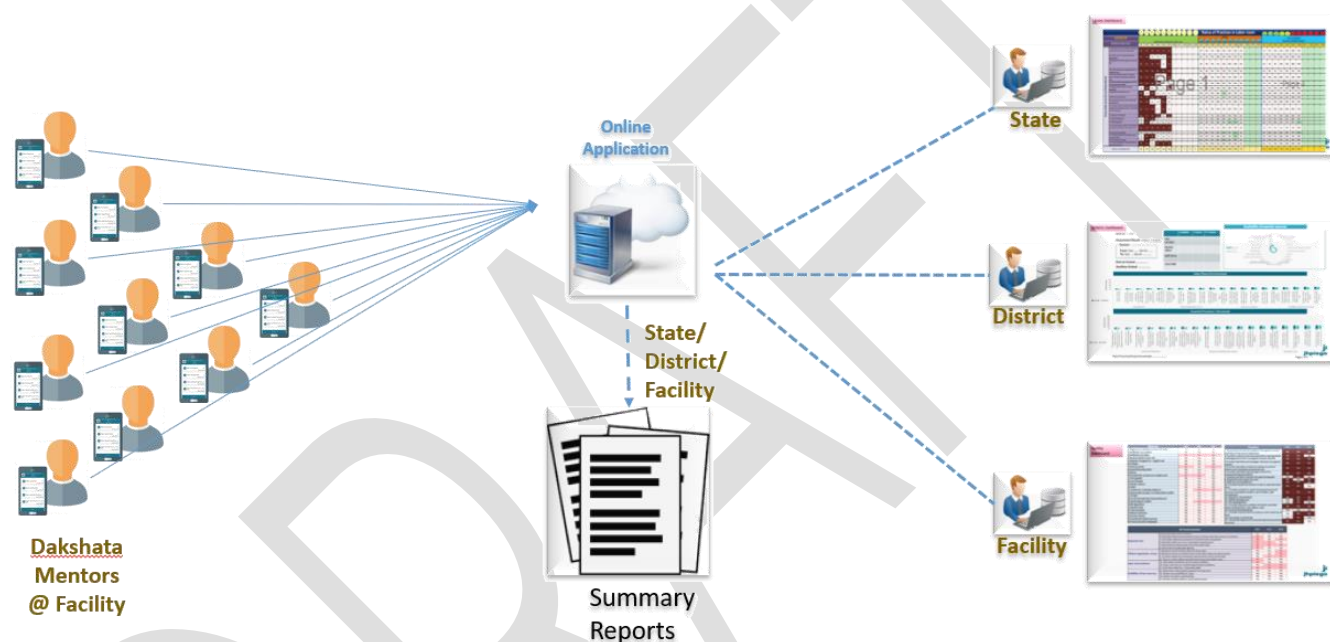


Fig 6- System Architecture of Dakshata Mentor App

Impact generated

- **Enhanced Facility Performance:** Through 241 visits across 131 facilities in 25 districts, this application has significantly improved facility performance by providing targeted mentoring and real-time data tracking.
- **User Engagement:** With a total of 165 users actively engaged, the application has created greater involvement in monitoring and improving healthcare facilities.
- **Data-Driven Decision-Making:** By offering facility-specific data on Labor room staff, resource gaps, Labor room environments, and practices, the application empowers healthcare providers to make informed decisions and take meaningful actions.
- **Mentoring Efficiency:** The app records mentoring areas, enabling mentors to assess progress and make necessary adjustments based on past visits, ultimately increasing mentoring efficiency.
- **Effective State-Level Oversight:** It has enabled the State MH division to closely monitor facility performance and mentor movements, ensuring a streamlined approach to healthcare improvement.

66. National Health Mission. (2021). *Dakshata Mentor Application*.

3. Swasthya Sewa Dapoon-Integrated MIS GIS System-Assam^{67,68}

Background

Since 2008, the state has implemented the Health Management Information System (HMIS) and, subsequently, Mother and Child Tracking System (MCTS) in 2011. However, it became evident that these systems did not capture all the required data. Moreover, these systems operated independently, lacking any interconnection. This resulted in the creation of information silos for data related to HR, ASHA activities, VHND monitoring, financial oversight, and health infrastructure development. Also, there was no comprehensive HR database, leading to the collection of separate district-based reports for each instance, often resulting in erroneous outcomes.

Steps taken by the State Government

To address the above-mentioned gaps, the National Health Mission in Assam initiated the phased development and implementation of the Integrated MIS GIS System, known as Swasthya Sewa Dapoon. This system encompasses the following features:

- **Unified Database Structure:** The system was designed with a single database structure, ensuring that all modules share the same master tables. This simplifies system integration and facilitates data sharing among different modules.
- **Single Data Entry:** Throughout the entire system, data is entered only once. Once data is captured in one module, it is automatically shared with other systems, eliminating the need for duplicate data entry and reducing the risk of errors.
- **Open-Source Technology:** The system was developed using open-source technology, promoting cost-efficiency and accessibility.
- **In-House Expertise:** NHM, Assam utilized its in-house capabilities to develop and host the system on NHM's own server.



Figure 7- Dashboard for Swasthya Sewa Dapoon.

Impact generated

- The complete HR database of the Health and Family Welfare Department got accessible with just one click after the implementation of this system.
- Salary disbursement for all NHM employees got centralized, leading to heightened employee motivation
- The online application module has streamlined the recruitment process, aiding in the efficient screening of eligible candidates.
- Online transfer and posting are available for both State health service and NHM employees, enabling real-time information updates.
- The system's transparency is evident through the public release of the HR database.
- Appointments are now made transparently based on merit through counselling, with immediate distribution of appointment letters.

67. National Health Systems Resource Centre. (2021). *National Summits on Good Replicable Practices & innovations in Public Healthcare Systems in India*.
68. National Health Mission. (2021). *Evidence to Action Towards UHC*.

Recommended Readings

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